

3. Terraform Backend Block

- ✓ Statefile should be in centralized location (Storage Account).
- ✓ Every user can use that statefile and they can create resources using that.
- ✓ We can use same code and create resources in different environments, but for that we should create different workspaces and use different statefiles.

✚ Below I have created storage account and stored statefile inside container

The image shows a Google search result for "terraform azure storage account container" and a Terraform code editor window.

Google Search Results:

- Search query: terraform azure storage account container
- Results: Terraform Registry, https://registry.terraform.io/docs/providers/azurerm/resources/storage_container.html, https://registry.terraform.io/providers/hashicorp/azurerm/latest/docs/resources/storage_container.html

Terraform Code Editor:

The code editor shows the Terraform configuration for an Azure storage account and container. The code is as follows:

```
main.tf Dev1 X terraform.tfstate main.tf Dev2
Dev1 > main.tf > ...
2 provider "azurerm" {
9   tenant_id      = "4a16e6d8-97ae-4ea2-8fb9-c9aa52ccc600"
10  subscription_id = "c4bc06e7-05d4-47ea-a63f-7a136548588c"
11 }
12
13 #resourcegroup
14 resource "azurerm_resource_group" "Urg01" {
15   name       = "Uday-RG01"
16   location   = "Central US"
17 }
18
19 #StorageAccount
20 resource "azurerm_storage_account" "store" {
21   name                        = "udaystorageaccount123"
22   resource_group_name        = azurerm_resource_group.Urg01.name
23   location                   = azurerm_resource_group.Urg01.location
24   account_tier               = "Standard"
25   account_replication_type   = "LRS"
26
27   tags = {
28     environment = "staging"
29   }
30 }
31
32 resource "azurerm_storage_container" "example" {
33   name                = "utfstate"
34   storage_account_id  = azurerm_storage_account.store.id
35   container_access_type = "blob"
36 }
37
```

Home > Resource Manager | Resource groups

Resource Manager | Resource groups

Default Directory

Search

+ Create Manage view

Resource Manager

- All resources
- Favorite resources
- Recent resources
- Resource groups**
- Tags
- Organization
- Tools
- Deployments
- Help

Uday-RG01

Resource group

How to manage changes with deployment tools? Generate Bicep code to duplicate this resource group. +1

Search

+ Create Manage view Delete resource group Refresh

Group by n

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
- Automation
- Help

Essentials

Resources Recommendations

Filter for any field...

Type equals all Location equals all Add filter

Name	Type
udaystorageaccount123	Storage account

Home > Resource Manager | Resource groups > Uday-RG01 > udaystorageaccount123

udaystorageaccount123 | Containers

Storage account

Search

+ Add container Upload Refresh Delete Change access level Restore containers Edit columns

Search containers by prefix

Only show active containers

Showing all 1 items

Name	Last modified	Anonymous access level	Lease state
utfstate	09/03/2026, 16:53:05	Blob	Available

Next configure BackendBlock from local to storage.

- For that we need below 4.

```

EXPLORER
TERRAZURE
  Dev1
    .terraform
    .terraform.lock.hcl
    main.tf
    resource.tf
    terraform.tfstate
    terraform.tfstate.backup
  Dev2
    .terraform
    .terraform.lock.hcl
    main.tf

Dev1 > main.tf > terraform > backend "azurerm" > abc key
37
38 #BackendBlock
39 terraform {
40   backend "azurerm" {
41     access_key      = "6Xp2uBLVF+4nTA1X8WNYitLI1j6Pi5gaAzOA1gEHr129koYUt9T79zyYNEdmuEC1aRk7RLaZunj+ASTQ/zs0g=="
42     storage_account_name = "udaystorageaccount123"
43     container_name    = "utfstate"
44     key               = "Udayprod.terraform.tfstate"
45   }
46 }
47
48

```

Home > Uday-RG01 > udaystorageaccount123



udaystorageaccount123 | Access keys ☆ ...

Storage account

Search

Partner solutions

Resource visualizer

Data storage

Security + networking

Networking

Front Door and CDN

Access keys

Shared access signature

Encryption

Microsoft Defender for Cloud

Data management

Set rotation reminder Refresh Give feedback

Access keys authenticate your applications' requests to this storage account. Keep your keys in a secure location like Azure Key Vault, and replace them often with new keys. The two keys allow you to replace one while still using the other.

Remember to update the keys with any Azure resources and apps that use this storage account.

[Learn more about managing storage account access keys](#)

Storage account name

udaystorageaccount123

key1 Rotate key

Last rotated: 09/03/2026 (0 days ago)

Key

6Xp2uBLVF+4nTA1X8WNYitLl1j6Pi5gaAzOA1gEHr129koYUt9T79zyYNEdmuEC1a...

Hide

Connection string

*****...

Show

✓ **Let's run terraform init, then it will ask for conformation to copy existing state to new backend.**

```
PS C:\TerrAzure> cd C:\TerrAzure\Dev1\
PS C:\TerrAzure\Dev1> terraform init
Initializing the backend...
Acquiring state lock. This may take a few moments...
Do you want to copy existing state to the new backend?
Pre-existing state was found while migrating the previous "local" backend to the
newly configured "azurerm" backend. No existing state was found in the newly
configured "azurerm" backend. Do you want to copy this state to the new "azurerm"
backend? Enter "yes" to copy and "no" to start with an empty state.

Enter a value:
```

✓ **After YES, above key created below in statefile.**

Home > Uday-RG01 > udaystorageaccount123 | Containers

utfstate

Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

+ Add Directory Upload Change access level Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

utfstate

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

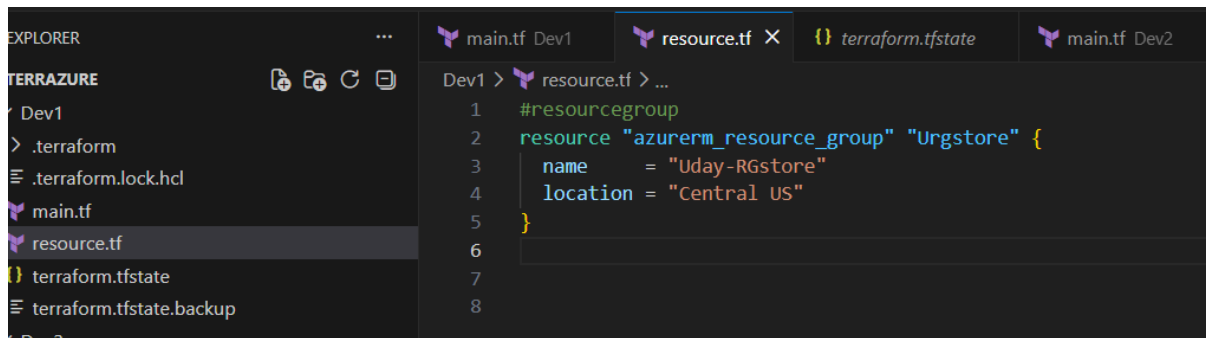
Search blobs by prefix (case-sensitive)

Only show active blobs

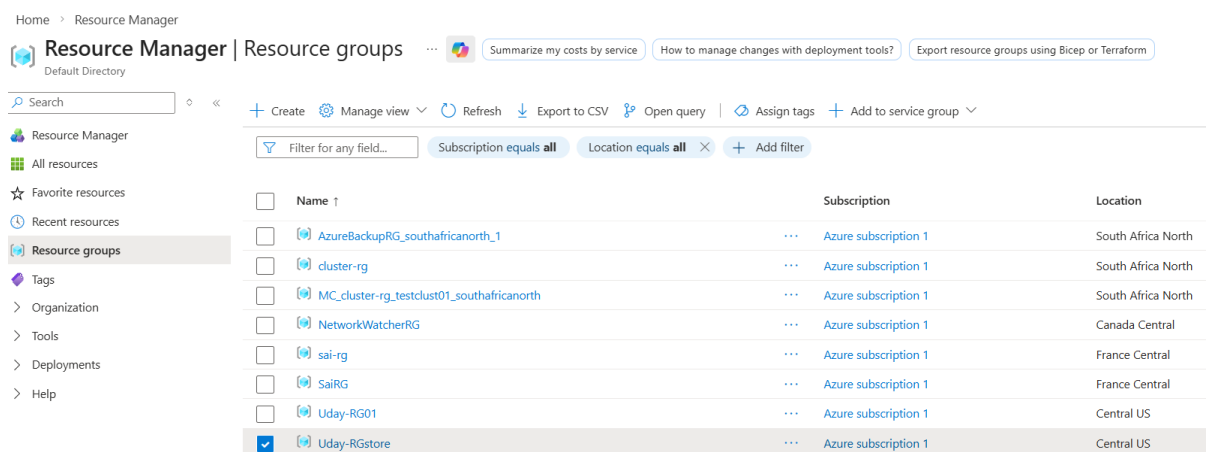
Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	Udayprod.terraform.tfstate	10/03/2026, 14:52:41	Hot (Inferred)	Block blob	12.45 KiB	Available

✓ Now let's create new RG,



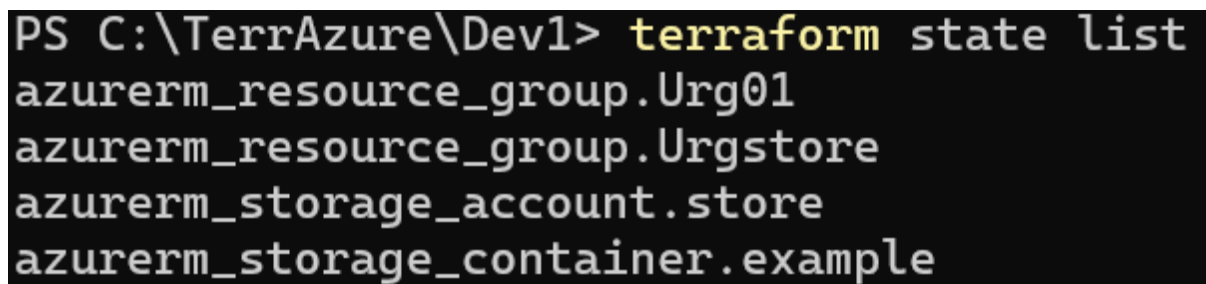
```
1 #resourcegroup
2 resource "azurerm_resource_group" "Urgstore" {
3     name     = "Uday-RGstore"
4     location = "Central US"
5 }
6
7
8
```



Name	Subscription	Location
AzureBackupRG_southafricanorth_1	Azure subscription 1	South Africa North
cluster-rg	Azure subscription 1	South Africa North
MC_cluster-rg_testclust01_southafricanorth	Azure subscription 1	South Africa North
NetworkWatcherRG	Azure subscription 1	Canada Central
sai-rg	Azure subscription 1	France Central
SaiRG	Azure subscription 1	France Central
Uday-RG01	Azure subscription 1	Central US
Uday-RGstore	Azure subscription 1	Central US

✓ Whatever resources created it will store in storage account statefile and disappear in local statefile.

✚ Below resources are available in statefile



```
PS C:\TerrAzure\Dev1> terraform state list
azurerm_resource_group.Urg01
azurerm_resource_group.Urgstore
azurerm_storage_account.store
azurerm_storage_container.example
```

✚ Above I want to remove particular resource from statefile but not from portal.

✚ Don't remove code from mainfile, if we do that the resources will delete from portal and from statefile.

✚ I don't want to delete any code, I just want to remove resources from statefile and delete other resource.

✓

```

PS C:\TerrAzure\Dev1> terraform state
Usage: terraform [global options] state <subcommand> [options] [args]

This command has subcommands for advanced state management.

These subcommands can be used to slice and dice the Terraform state.
This is sometimes necessary in advanced cases. For your safety, all
state management commands that modify the state create a timestamped
backup of the state prior to making modifications.

The structure and output of the commands is specifically tailored to work
well with the common Unix utilities such as grep, awk, etc. We recommend
using those tools to perform more advanced state tasks.

Subcommands:
  identities      List the identities of resources in the state
  list            List resources in the state
  mv             Move an item in the state
  pull           Pull current state and output to stdout
  push           Update remote state from a local state file
  replace-provider Replace provider in the state
  rm             Remove instances from the state
  show           Show a resource in the state
PS C:\TerrAzure\Dev1>

```

```

PS C:\TerrAzure\Dev1> terraform state list
azurerm_resource_group.Urg01
azurerm_resource_group.Urgstore
azurerm_storage_account.store
azurerm_storage_container.example

```

- ✓ Let's remove container & storage,Rg-store (to remove dependency for storage account) account from statefile.

```

PS C:\TerrAzure\Dev1> terraform state rm azurerm_storage_container.example
Acquiring state lock. This may take a few moments...
Removed azurerm_storage_container.example
Successfully removed 1 resource instance(s).
Releasing state lock. This may take a few moments...

```

```

PS C:\TerrAzure\Dev1> terraform state rm azurerm_storage_account.store
Acquiring state lock. This may take a few moments...
Removed azurerm_storage_account.store
Successfully removed 1 resource instance(s).
Releasing state lock. This may take a few moments...

```

```

PS C:\TerrAzure\Dev1> terraform state rm azurerm_resource_group.Urgstore
Acquiring state lock. This may take a few moments...
Removed azurerm_resource_group.Urgstore
Successfully removed 1 resource instance(s).
Releasing state lock. This may take a few moments...

```

- ✓ After removing, we have only 1 below.

```
PS C:\TerrAzure\Dev1> terraform state list
azurerm_resource_group.Urg01
```

- ✚ Now if we **remove above 3 resources** from **code** and when we run **terraform apply**, it will show **infra-matches** below result.

```
PS C:\Users\kucha\Desktop\Azureterra\Dev1> terraform apply
Acquiring state lock. This may take a few moments...
azurerm_resource_group.rg01: Refreshing state... [id=/subscriptions/0cf71a42-7c78-413c-aa9a-e3adaf9c9972/resourceGroups/standard-rg]
azurerm_virtual_network.vnet01: Refreshing state... [id=/subscriptions/0cf71a42-7c78-413c-aa9a-e3adaf9c9972/resourceGroups/standard-rg/providers/Microsoft.Network/virtualNetworks/vnet0111]

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes are needed.
```

❖ ACQUIRING STATE LOCK

```
PS C:\Users\kucha\Desktop\Azureterra\Dev2> terraform destroy
Acquiring state lock. This may take a few moments...
azurerm_resource_group.rg01: Refreshing state... [id=/subscriptions/0cf71a42-7c78-413c-aa9a-e3adaf9c9972/resourceGroups/standard-rg]
```

- When 2dev are working, this acquiring state lock will be applied one after another, so that only one dev will work at a time.